

DSA - Assignment - 4

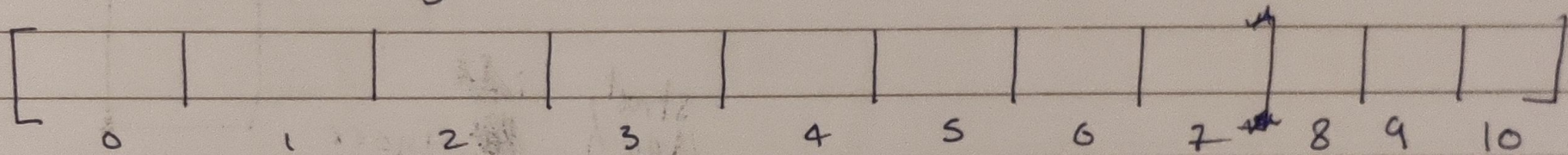
Q.1

Given:- Hash table of size $m = 11$

$\rightarrow$ 10, 22, 34, 4, 15, 28, 17, 88, 59

Given key to be inserted in Hash table

(a) Linear probing $h(k, i) = (k + i) \bmod m$



$\rightarrow$ 1. Key = 10

$i = 0$, $(10 + 0) \% 11 = 10$, No Collision
 $\uparrow$ index of insertion

$\rightarrow$ 2. Key = 22

$i = 0$, $(22 + 0) \% 11 = 0$, No collision
 $\uparrow$ index of insertion

$\rightarrow$ 3. Key = 34

$i = 0$, $(34 + 0) \% 11 = 1$, No Collision
 $\uparrow$ index of insertion

$\rightarrow$ 4. Key = 4

$i = 0$, $(4 + 0) \% 11 = 4$, No Collision
 $\uparrow$ index of insertion

$\rightarrow$ 5. Key = 15

$i = 0$, $(15 + 0) \% 11 = 4$, Collision! taken by Key = 4

$i = 1$, $(15 + 1) \% 11 = 5$, No collision, '5' is the index

6. Key = 28

$i=0$, $(28+0) \% 11 = \textcircled{6}$, No Collision
 $\uparrow$ → index of insertion

7. Key = 17

$i=0$, $(17+0) \% 11 = 6$, Collision!! ~~6~~ taken by Key = 28

$i=1$, $(17+1) \% 11 = \textcircled{7}$ ← No Collision, 7 is the index
~~6~~

8. Key = 88

$i=0$, $(88+0) \% 11 = 0$ collision, taken by '22'

$i=1$, $(88+1) \% 11 = 1$ collision, taken by '34'

$i=2$, $(88+2) \% 11 = \textcircled{2}$ NO collision, 2 is the index

9. Key = 59

$i=0$, $(59+0) \% 11 = 4$ collision, taken by '4'

$i=1$, $(59+1) \% 11 = 5$ collision, taken by '15'

$i=2$, $(59+2) \% 11 = 6$ collision, taken by '28'

$i=3$, $(59+3) \% 11 = 7$ collision, taken by '17'

$i=4$, $(59+4) \% 11 = \textcircled{8}$ NO collision, 8 is the index

(a) final Hash table

Index Value

0	22
1	34
2	88
3	-
4	4
5	15
6	28
7	17
8	59
9	-
10	10

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Q1] (b) Double Hashing

$$h_1(K) = K \bmod m,$$

$$h_2(K) = 1 + (K \bmod (m-1))$$

Probing function together

1.

$$h(K, i) = (h_1(K) + i \cdot h_2(K)) \bmod 11$$

1. insert Key = 10

$$i=0, 10 \cdot 11 = 10$$

↑
~~index~~

Placed at index 10

2. insert Key = 22

$$i=0, 22 \cdot 11 = 0$$

↑

Placed at index 0

3. insert Key = 34

$$i=0, 34 \cdot 11 = 1$$

↑

Placed at index 1

4. insert Key = 4

$$i=0, 4 \cdot 11 = 4$$

↑

Placed at index 4

5, insert Key = 15

$$\sim i=0 \quad 15 \times 11 = 4$$

↑

collision by 4

$$\sim i=1,$$

$$h_1(15) = 15 \times 11 = 4$$

$$h_2(15) = 1 + (15 \times 10) = 6$$

$$\rightarrow (4 + 6) \times 11 = 10$$

↑

collision by 10

$$\sim i=2,$$

$$h_1(15) = 4$$

$$h_2(15) = 1 + (15 \times 6)$$

$$\rightarrow (4 + 2 \times 6) \times 11 = 5$$

↑

placed at index 5

6) insert Key = 28

$$i=0, \quad h_1(28) = 28 \times 11 = 6$$

↑

Placed at index $\rightarrow 6$

7) insert Key = 17

$$i=0,$$

$$17 \times 11 = 6$$

↑

collision
by 28

$$i=1, \quad h_1(17) = 6$$

$$h_2(17) = 1 + (17 \times 10) = 8$$

$$\rightarrow (6 + 1 \times 8) \times 11$$

$$\rightarrow 3$$

↑

placed at index

insert Key = 88

$$i=0, h_1(88) = 88 \cdot 11 = 0$$

↑

collision by 22

$$i=1, h_1(88) = 0$$

$$h_2(88) = 1 + (88 \cdot 10) = 9$$

$$\rightarrow (0 + 1 \times 9) \cdot 11$$

$$\Rightarrow 9$$

↑

Placed at index 9

insert Key = 59

$$i=0,$$

$$h_1(59) = 59 \cdot 11 = 4$$

↑

collision by
'4'

$$i=1, h_1(59) = 4$$

$$h_2(59) = 1 + (59 \cdot 10) = 10$$

$$\rightarrow (4 + 10) \cdot 11 = 2$$

↑

collision
by 2

$$i=2, h_1(59) = 4$$

$$h_2(59) = 10$$

$$\rightarrow (4 + 2 \times 10) \cdot 11 = 2$$

Placed at index '2'

(b)

final HashTable

0 22

1 34

2 59

3 17

4 4

5 15

6 28

7 -

8 -

9 88

10 10

Q3 (a) pseudocode for MIN-HEAPIFY

MIN-HEAPIFY(A, i)

$n = A.\text{heap-size}$

$L = 2 \times i$

$R = 2 \times i + 1$

$\text{smallest} = i$

~~if $L \leq n$ and $A[L] < A[\text{smallest}]$~~

if $L \leq n$ and $A[L] < A[\text{smallest}]$

$\text{smallest} = L$

if $R \leq n$ and $A[R] < A[\text{smallest}]$

$\text{smallest} = R$

if $\text{smallest} \neq i$

Swap $A[i] \leftrightarrow A[\text{smallest}]$ # Swap these two elements

MIN-HEAPIFY(A, smallest)

(b) Given:- $A = [10, 3, 5, 7, 2, 8]$ initially

1. MIN-HEAPIFY(A, 1), $i = 1$

$L = A[2] = 3$, $R = A[3] = 5$

$\text{smallest} \neq i$, $A[1] \leftrightarrow A[2]$

$\rightarrow$ Heap MIN-HEAPIFY(A, 2)

2. ~~1st~~ $L = A[4] = 7$, $R = A[5] = 2$

$\text{smallest} = 5$, why? $\text{smallest} \neq 2 \rightarrow$ Swap $A[2]$ and $A[5]$

84 3. MIN-HEAPIFY(A, 5)

L = NO children, R = NO children

No Sw Smallest = 5

So, NO Recursive Call,

final MIN-HEAP ~~array~~ array

$$\rightarrow A = [2, 3, 5, 7, 10, 8]$$

Q. 11)

$\rightarrow$ MIN-HEAPIFY and Max-HEAPIFY both have identical running time $\sim O(n \log n)$

Why? Because both algorithms are structurally same they only differ in the direction of operation operator ' $<$ ' & ' $>$ '

The work done each level is $O(1)$, Compare a node against its two children, identify minor Max and Swap if needed. Then recursion depends to one level into one subtree.

A heap over ' n ' elements has height ' $\log n$ ', So worst case root to leaf path will be taken $T(n) = O(n \log n)$.